

AYUBA GARBA

Systems & Software Engineer | Distributed Architectures & Algorithmic Engines

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EXECUTIVE ENGINEERING SUMMARY

Systems & Software Engineer specializing in resilient, high-concurrency distributed backends (Go, Rust), algorithmic execution engines (MQL5, C++), and cloud infrastructure. Track record architecting zero-downtime reverse proxies, deterministic row-locking ledgers with strict ACID guarantees, and sub-millisecond market routing. Vetted by Turing (Top 1%) and certified across AWS Cloud AI and Walmart Global Tech software engineering tracks.

CORE TECHNICAL STACK & ARCHITECTURE MATRIX

Systems & Core Languages:	Go (Golang), Rust, C++, MQL5, TypeScript, JavaScript (ESNext), Python, SQL
Distributed Systems & Concurrency:	Non-blocking I/O, Goroutine Worker Pools, Mutexes, WebSockets Multiplexing, Redis Pub/Sub
State Invariance & Databases:	PostgreSQL, TimescaleDB, Redis, SQLite, MongoDB, ACID Transactions, Row-Level Locking
Cloud, Edge & Reliability:	Cloudflare Edge Workers, AWS, Linux Kernel Tuning, Docker, CI/CD Automation, Netlify
Security & Algorithmic Engines:	zk-SNARK Biometric Circuits, RBAC Access Matrices, 3-EMA Market Pullbacks, Dynamic ATR

KEY ARCHITECTURAL SYSTEMS & PRODUCTION DEPLOYMENTS

NeXaVerSe MVP — Distributed Super-App & Zero-Knowledge Identity Engineered an asynchronous Go reverse-proxy API gateway with non-blocking worker pools achieving <1.2ms routing latency. Designed a double-entry ACID Rust ledger implementing deterministic account row-locking to mathematically prevent deadlock conditions. Multiplexed real-time WebSocket chat and session presence across horizontal Redis pub/sub nodes with zero message drops. Integrated zero-knowledge biometric gating circuits (zk-SNARKs) verifying identity credentials without raw biometric disclosure.	Go Rust zk-SNARKs WebSockets Redis Cluster
Automated FX Execution Engine — MetaTrader 5 Institutional Engine Engineered sub-millisecond algorithmic execution engine in MQL5 and C++ processing high-frequency tick data streams. Built 3-EMA structural pullback detection combined with dynamic ATR volatility position sizing to enforce strict risk controls. Deployed stepped trailing stops and automated breakeven lock logic to eliminate downside market slippage on active corridors. Integrated multi-timeframe RSI momentum validation and automated hardware circuit-breaker kill switch for abnormal spread anomalies.	MQL5 C++ Engine 3-EMA Strategy TimescaleDB Sub-millisecond Ticks
KingdomConnect — Scalable Multi-Tenant Collaborative Fabric Architected collaborative communication platform supporting concurrent multi-tenant community nodes with low-latency delivery. Designed granular role-based access control (RBAC) matrix isolating sensitive administrative domains and organizational scopes. Synchronized horizontal message broadcasts across distributed nodes using Redis pub/sub channel clustering.	TypeScript Distributed Redis PostgreSQL RBAC Matrix
Production Deployed Client Platforms (Awsoli, Taila, SaviLinks, Defenzio) Awsoli (awsoli.org): Global Cloudflare edge asset caching, sub-second routing, and responsive client typography with 99.99% uptime. Taila (taila.app): High-throughput digital product platform with sub-150ms interaction corridors and authenticated API session gateways. SaviLinks & Defenzio: Built enterprise message dispatch pipelines with DLQ retry logic and hardened Linux defense postures.	Angular Go Node.js/TypeScript PostgreSQL Hardened Linux

VETTED CREDENTIALS & PROFESSIONAL ACCREDITATIONS

Advanced Software Engineering — Walmart Global Tech (Forage) Evaluated in Java/Python microservices, relational query optimization, distributed caching, and enterprise CI/CD.	2024
AWS Cloud AI Practitioner — Udacity / AWS (Verified Track) Evaluated on cloud-native AI pipeline deployment, sub-millisecond inference integration, and AWS security best practices.	2024
Emotional Intelligence & Leadership — University of the People Accredited coursework in high-performance team leadership, emotional governance, and cross-functional engineering execution.	2024
Turing Vetted Systems & Software Engineer Vetted across algorithmic depth, concurrent systems design, systems scalability, and production code quality.	Top 1% Global